Sentiment Classification using LSTM-21/10/25(Tuesday):

Project Summary & Performance Report

This project successfully implemented a Long Short-Term Memory (LSTM) neural network to perform binary sentiment classification on the IMDb Movie Reviews dataset.

The model was trained for 10 epochs using word embeddings learned from scratch. The primary goal was to classify movie reviews as either Positive (1) or Negative (0).

Key Results & Observations

1. Data and Preprocessing

- Dataset: IMDb Movie Reviews (50,000 total reviews).
- Class Balance: The dataset is perfectly balanced (25,000 Negative, 25,000 Positive), so no imbalance handling was required.
- Max Sequence Length (MAX_LEN): Statistical analysis determined the 95th percentile length was 598 tokens. All sequences were padded/truncated to this length.

2. Model Performance (Test Set)

The model's final performance on the unseen Test Set (25,000 reviews) was very poor, performing close to random chance.

Metric	Score	Interpretation
Test Accuracy	0.5024	The model correctly classified only 50.24% of the reviews.
Test Loss	0.7678	The loss is high, confirming poor performance.
F1-Score (Positive Class)	0.0635	Extremely low. The model failed to correctly identify positive reviews.

3. Deep Dive into Classification Issues

The Classification Report and Confusion Matrix reveal a severe problem: Class Imbalance in Prediction.

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Class	Precision	Recall	F1-Score	Support	
Negative (0)	0.5	0.97	0.66	12500	
Positive (1)	0.54	0.03	0.06	12500	
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? Model Bias: The model learned to predict the Negative (0) class most of the time.

- It achieved a high Recall (0.97) for the Negative class, meaning it correctly identified almost all the actual negative reviews.
- It achieved an extremely low Recall (0.03) for the Positive class, meaning it missed almost all the actual positive reviews.

? Conclusion: The model essentially defaulted to predicting "Negative" for nearly every review, leading to a $\approx 50\%$ accuracy simply by guessing one class consistently.

4. Training Progress (Loss/Accuracy Curves)The training logs confirm the model failed to learn effective patterns:Training Loss: Hovered around $\mathbf{0.65}$ to $\mathbf{0.69}$ and was often higher than the validation loss in early epochs, but then the validation loss soared.Validation Loss: The Validation Loss continually increased (from 0.6928 to 0.7834) while the Training Accuracy remained near $\mathbf{0.50}$.Observation: This pattern of soaring validation loss is a classic sign of extreme overfitting or a vanishing/exploding gradient problem in the LSTM, causing the model's weights to become unstable and fail to generalize even slightly.